AI & Data Science Cell · ENSIA Tech Community

A transformer at a glance

GPT-2 small, with every tensor shape for a sequence of T tokens. (Sessions 0 to 3 and 9 of the AI Journey advanced course.)

Forward pass

token ids	`(T)`	integers from 0 to 50,256
token embedding + position embedding	`(T, 768)`	50,257 x 768 table (38.6M params) + a vector per position
layer norm	`(T, 768)`	a normalised copy; the block's residual stream is left alone
q, k, v from one matrix (768 x 2304)	`3 x (T, 768)`	split into 12 heads: (12, T, 64) each
scores q k^T / sqrt(64)	`(12, T, T)`	causal mask: later positions set to minus infinity
softmax over the last axis	`(12, T, T)`	each row sums to 1; the first row reads only itself
weights x v, heads joined, output matrix 768 x 768	`(T, 768)`	added back to the residual stream
layer norm, then MLP 768 -> 3072 -> 768	`(T, 3072) -> (T, 768)`	applied to each position on its own; added back
12 blocks like this, then a final layer norm	`(T, 768)`	the logit lens reads the stream after any block
output layer (the embedding table again, tied)	`(T, 50,257)`	one logit per token; softmax gives probabilities

One block, in formulas (pre-norm)

```
x' = x + Attn( LN(x) )
y  = x' + MLP( LN(x') )
Attn = softmax(Q K^T / sqrt(d_k) + mask) V
per block: attention 2,362,368
          MLP 4,722,432 (two thirds)
12 blocks hold about 85M of 124M
```

Numbers worth knowing

GPT-2 small: 12 layers, 12 heads of 64, d = 768

Training compute is about 6 N D operations

KV cache per token = 2 x layers x kv heads

x head size x bytes

GPT-2 36,864 B; Qwen2.5-0.5B 12,288 B (16-bit)

Bits per token = -log2 p; perplexity = 2^mean bits

Three layouts, one block (Session 4)

encoder (BERT)	no mask: every token reads every token; trained to recover masked tokens
decoder (GPT-2)	causal mask: a token reads itself and earlier tokens; trained to predict the next token
encoder-decoder (T5)	encoder without a mask, decoder with one, plus cross-attention from target to source

Sampling: greedy takes the most likely token; temperature divides the logits; top-k keeps the k most likely tokens;

top-p keeps the smallest set of tokens whose probabilities add up to p. Decoding costs one forward pass per new token.

Before you trust a number (Sessions 10 and 11)

- Report an interval: about plus or minus 3 points on 1,000 questions. Compare models on the same items.
- Several seeds, a paired interval for the difference, and new seeds for the winner of a search.
- Count your comparisons: 20 tweaks that change nothing gave a 'significant' one in 28% of simulated studies.
- Check a judge in both answer orders; check the evaluation data for overlap with the training data.

Source: the notebooks of the AI Journey advanced course (third edition). Every number the course quotes is printed by its notebooks.